



MODELING AND CONTROL OF BACKLASH AND FRICTION IN MANIPULATORS

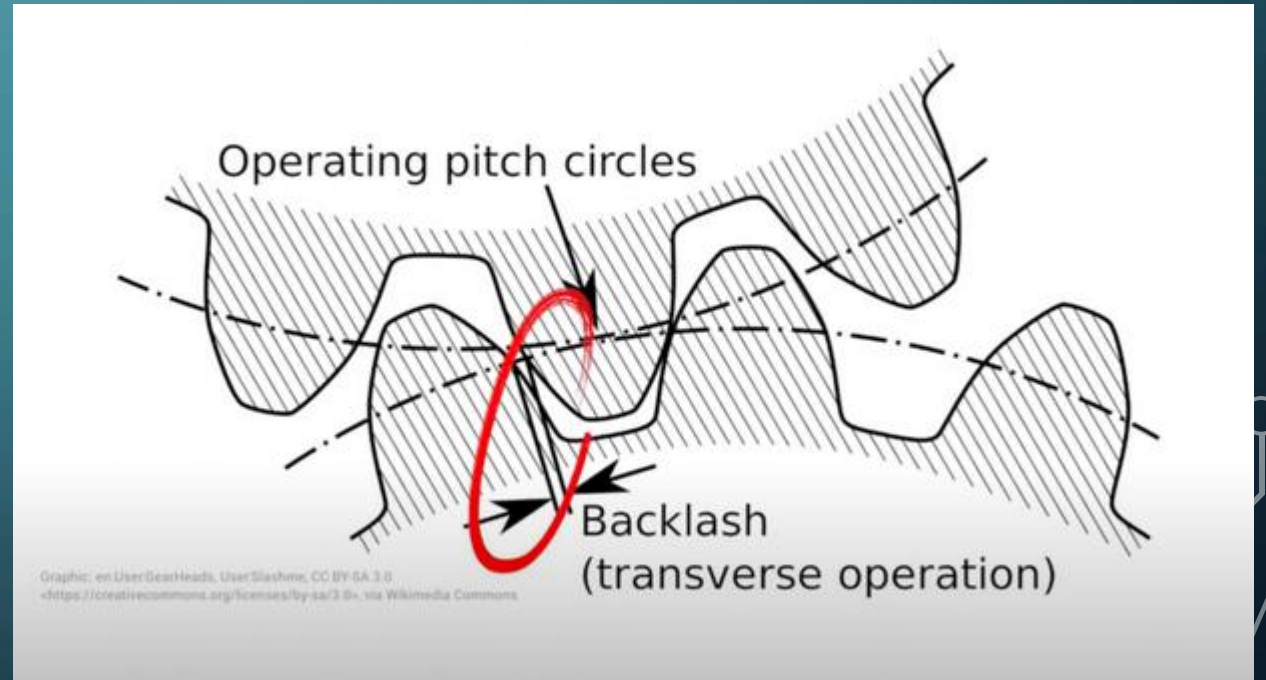
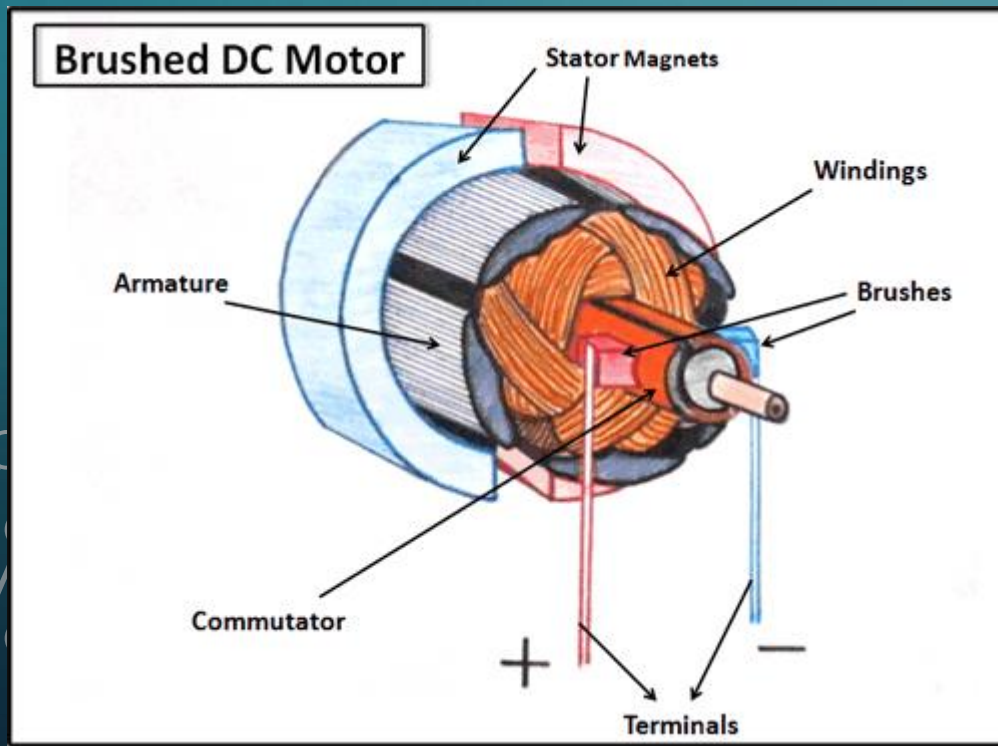
ANNA BURGESS

JUSTICE HICKMAN-MAYNARD

ES-259 – INTRODUCTION TO ROBOTICS S26

BACKGROUND

In robotics applications, the dominating non-linearities are friction and backlash.



WHY THIS MATTERS?

- Motors typically use linearized dynamic control models that do not account for these nonlinearities.

$$\tau = M(\theta)\ddot{\theta} + C(\theta, \dot{\theta})\dot{\theta} + g(\theta)$$

$$\tau = K_p\theta_e + K_i \int \theta_e(t)dt + K_d\dot{\theta}_e$$

$$\tau = \tilde{M}(\theta) \left(\ddot{\theta}_d + K_p\theta_e + K_i \int \theta_e(t)dt + K_d\dot{\theta}_e \right) + \tilde{h}(\theta, \dot{\theta})$$

PROPOSED MODEL:

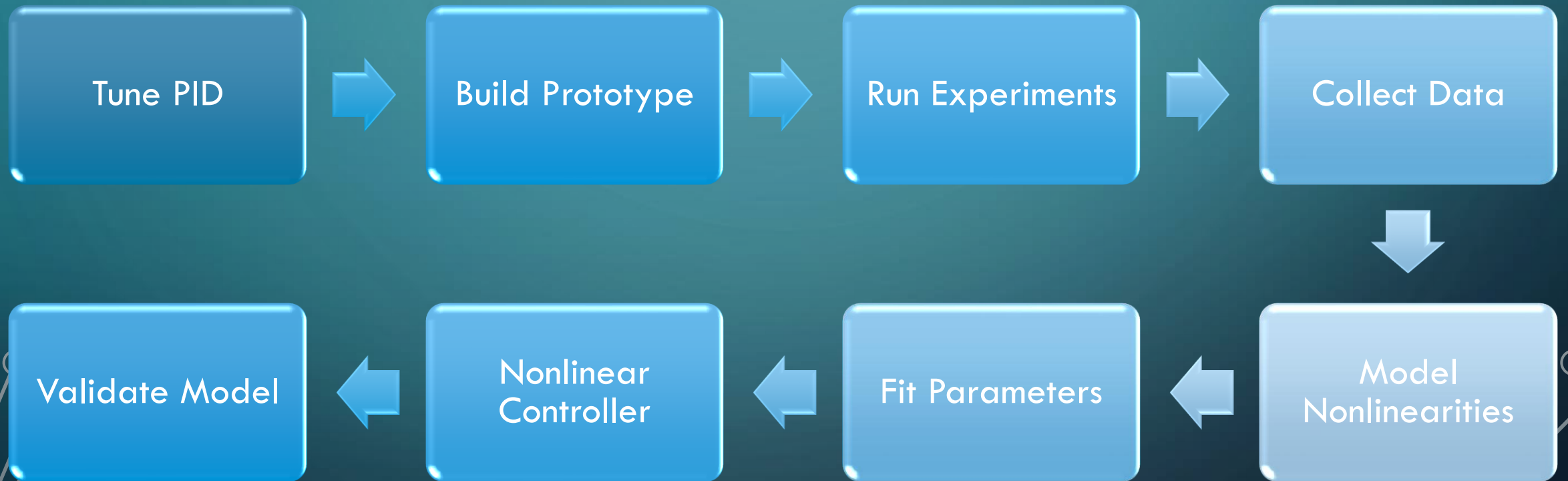
- We extend computed torque control by adding friction feedforward in the motor dynamics and compensating backlash through a nonlinear mapping between desired output and motor-side motion.

$$\tau = \tilde{M}(q_m)(\ddot{q}_{m,d} + K_d \dot{e}_m + K_p e_m + K_i \int e_m dt) + \tilde{h}(q_m, \dot{q}_m) + \tilde{\tau}_f(\dot{q}_m)$$

$$\tau_f = \tau_{\text{output}} - J\alpha$$

$$q_m = q_d + b$$

METHODOLOGY



PID TUNING

Chosen Values

Gain

Value

K_p

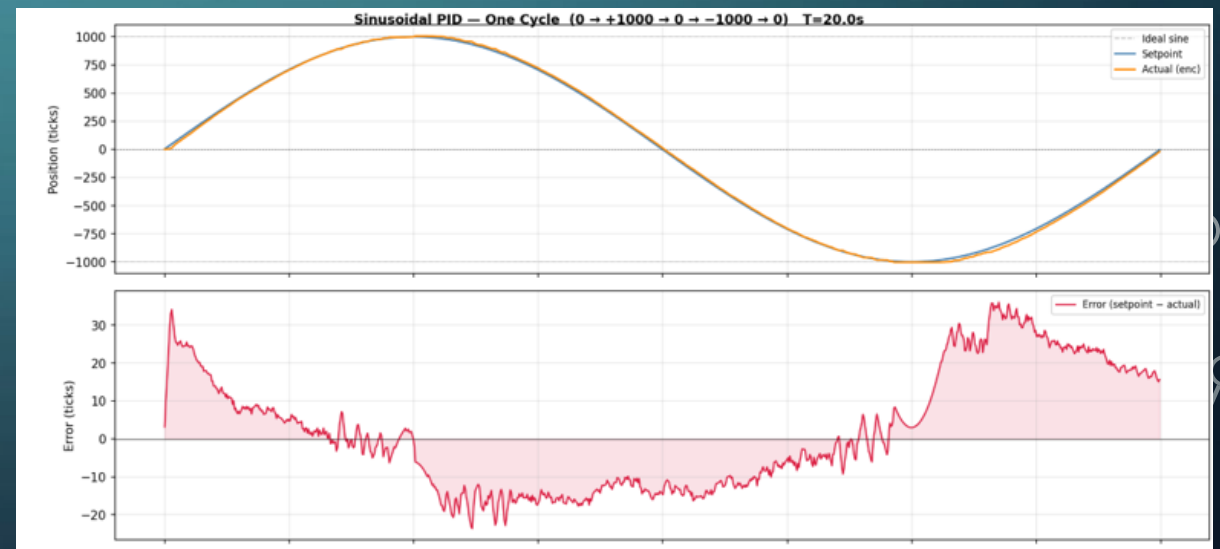
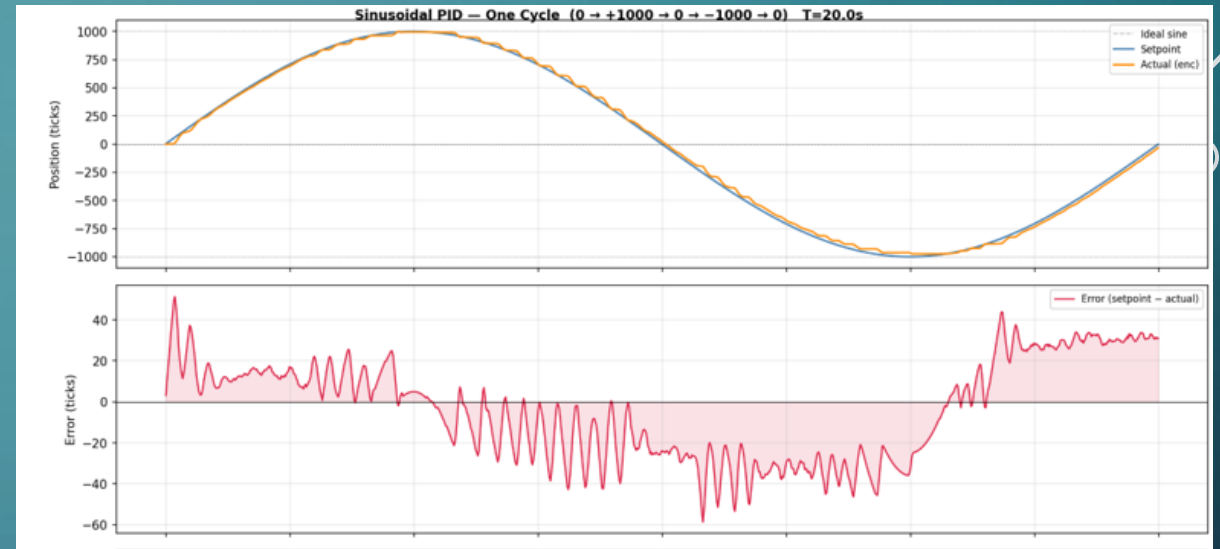
1.5

K_i

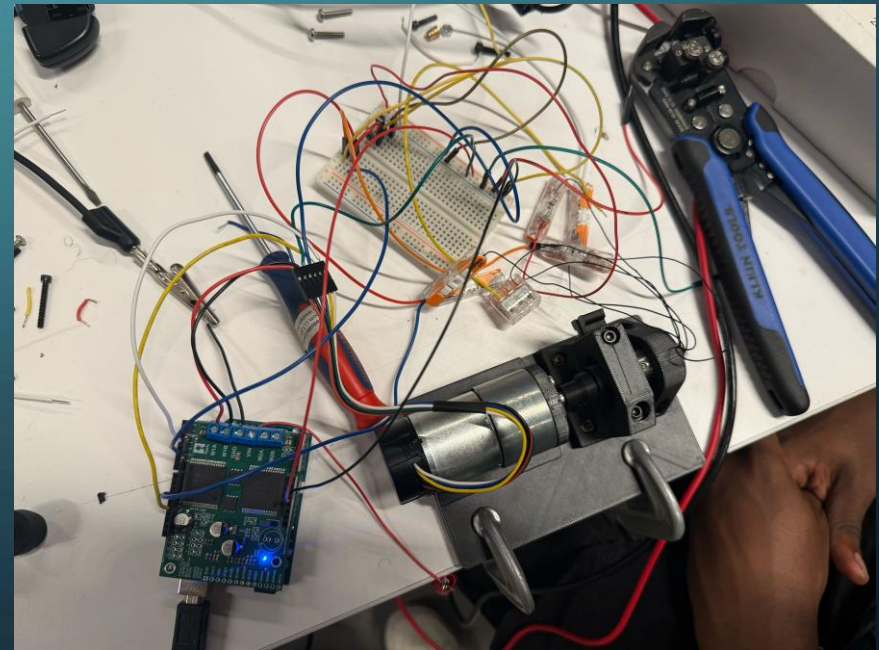
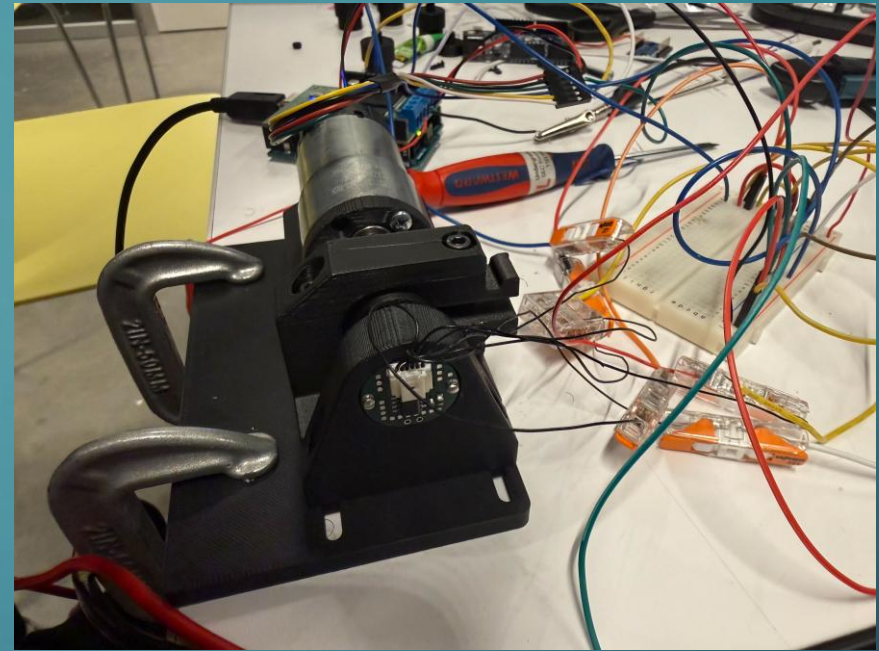
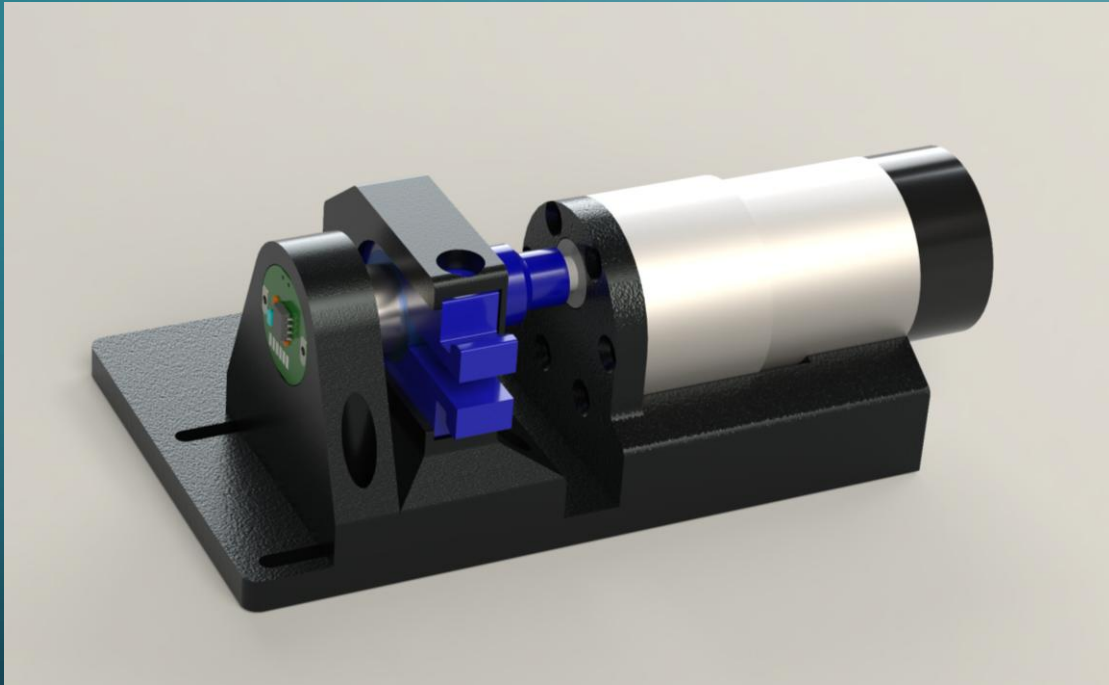
0.5

K_d

0.05

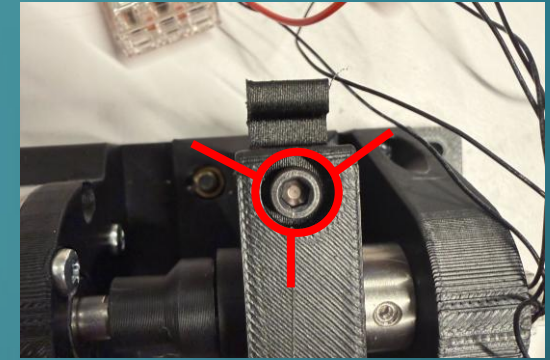


BUILD PROTOTYPE

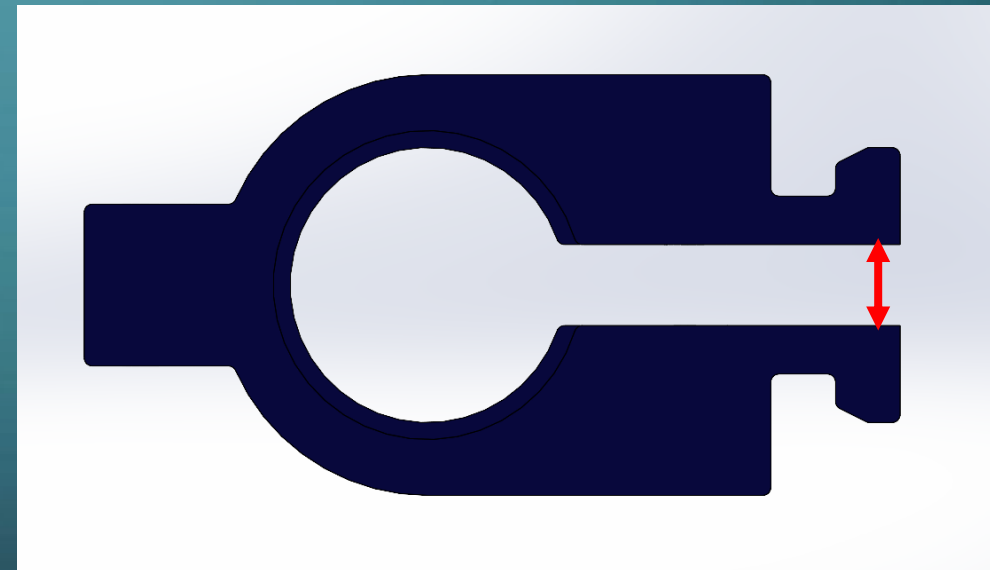


FRICTION INDUCER

- 4 Levels of Friction:
 - 1st level was least friction (backed all the way out). We scored the screw and marked a position on the print.
 - 4th level was most friction (backed in until tug (DIDN'T TORQUE). This was two full rotations.
 - 2nd and 3rd level were at $2/3$ rotation and $4/3$ rotation, respectively.



Tightening Screw Here

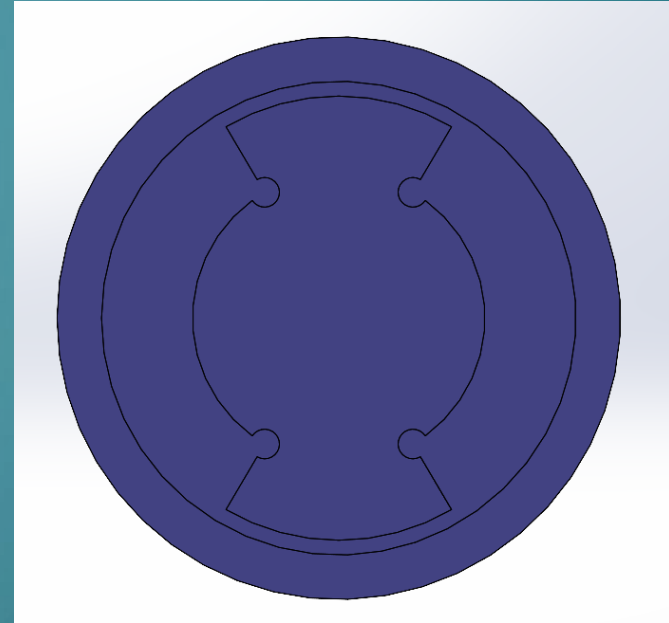


↓ Distance = ↑ Friction

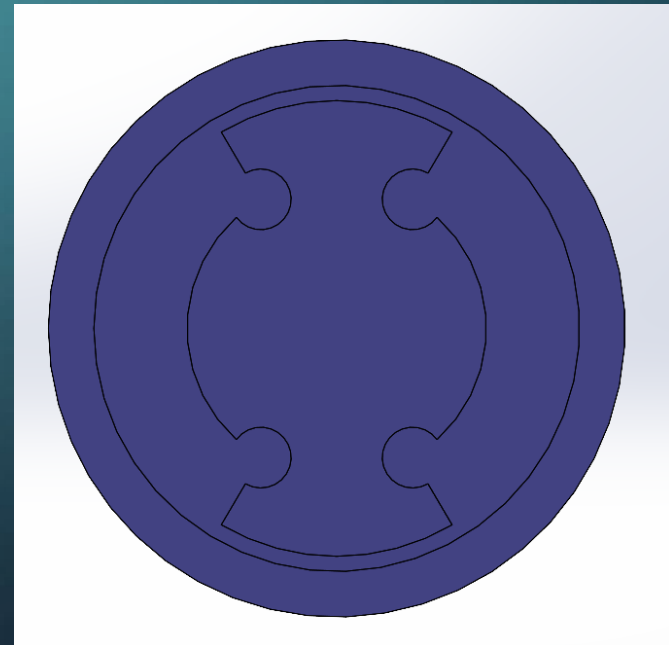
BACKLASH INDUCER

- 5 Levels of Backlash:
- Began at “Default” and increased in increments of 0.1 mm in radius for backlash per inducer.
- Default
- 0.1 mm
- 0.3 mm
- 0.4 mm
- 0.5 mm

Largest Backlash

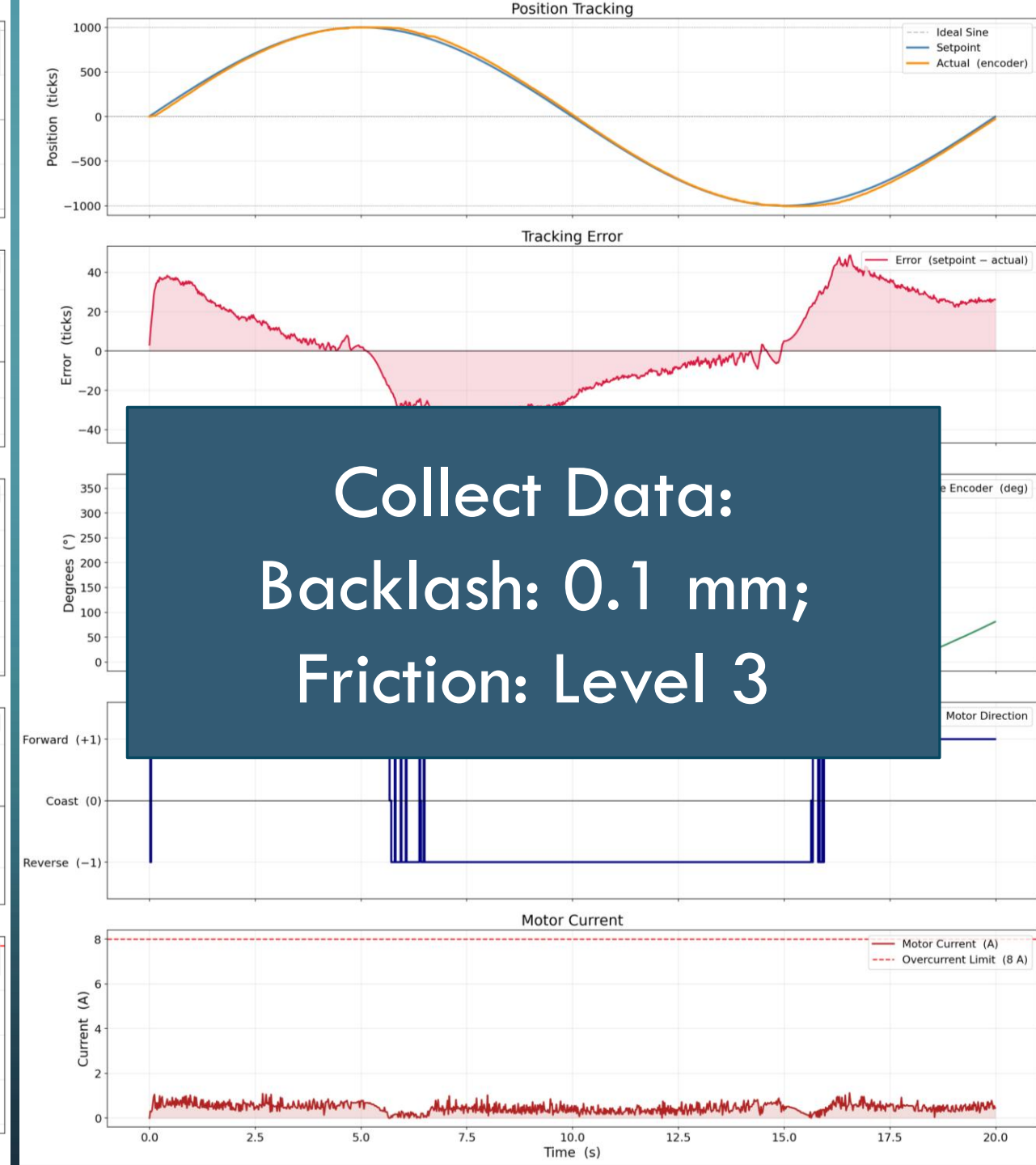
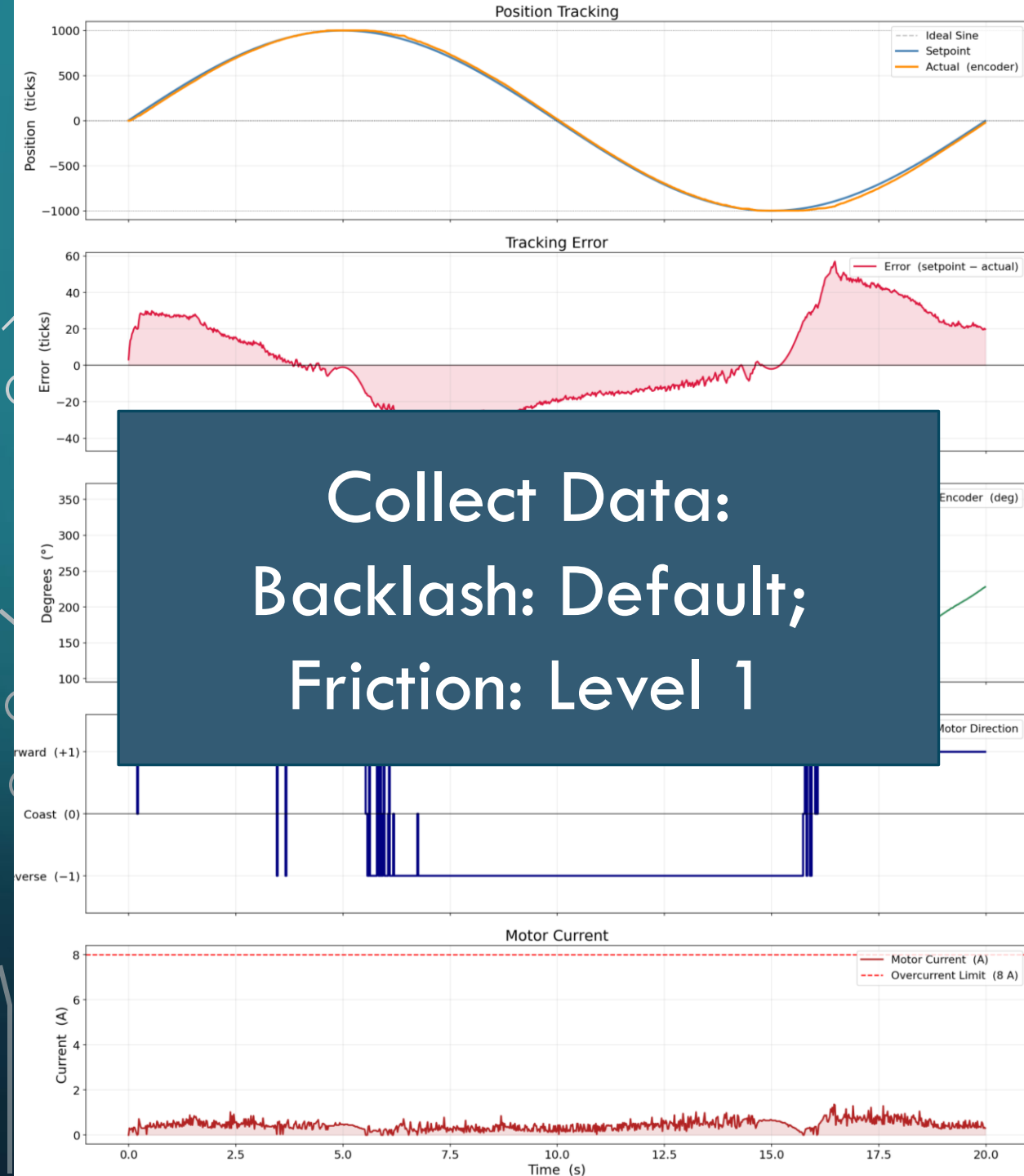


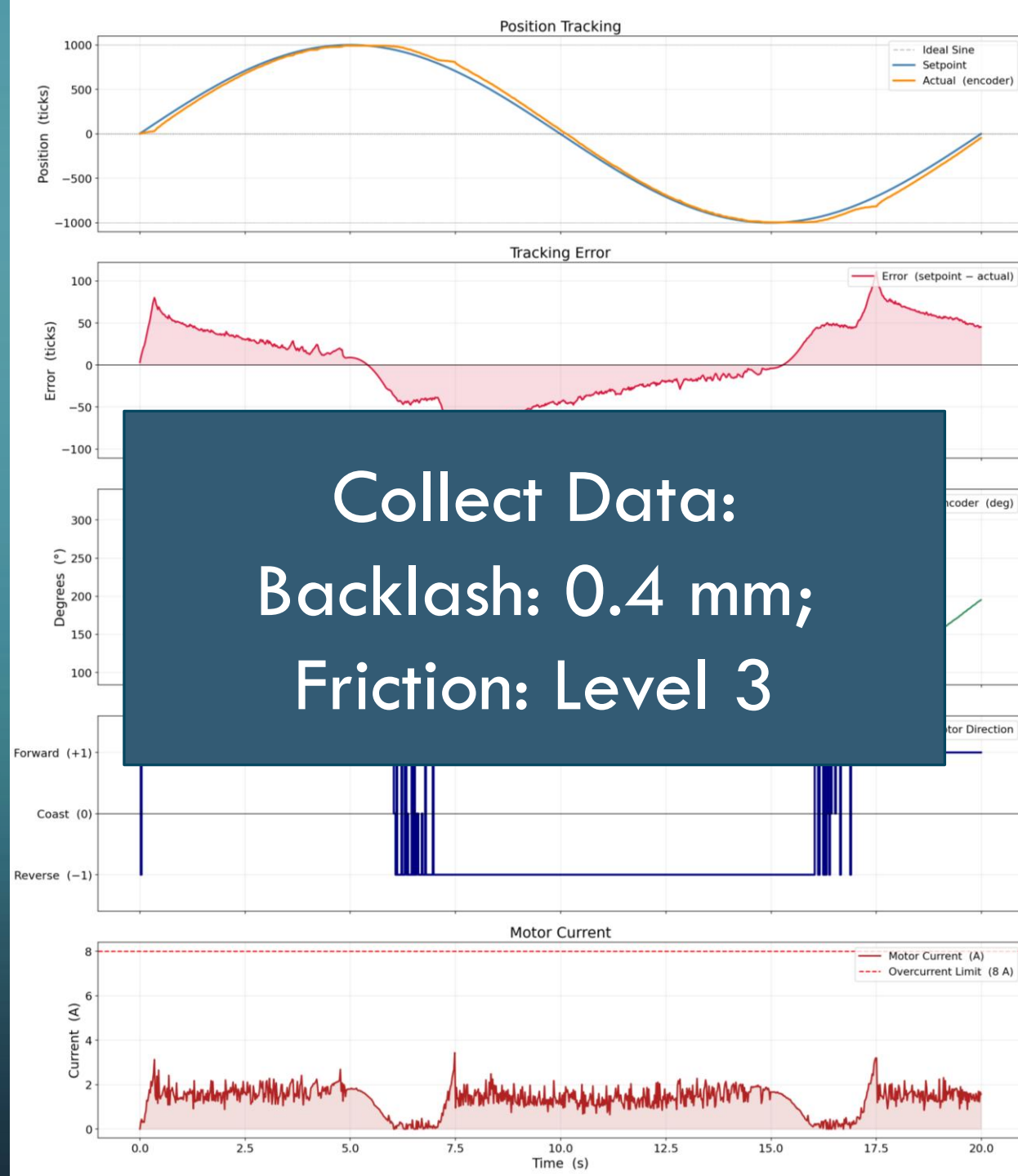
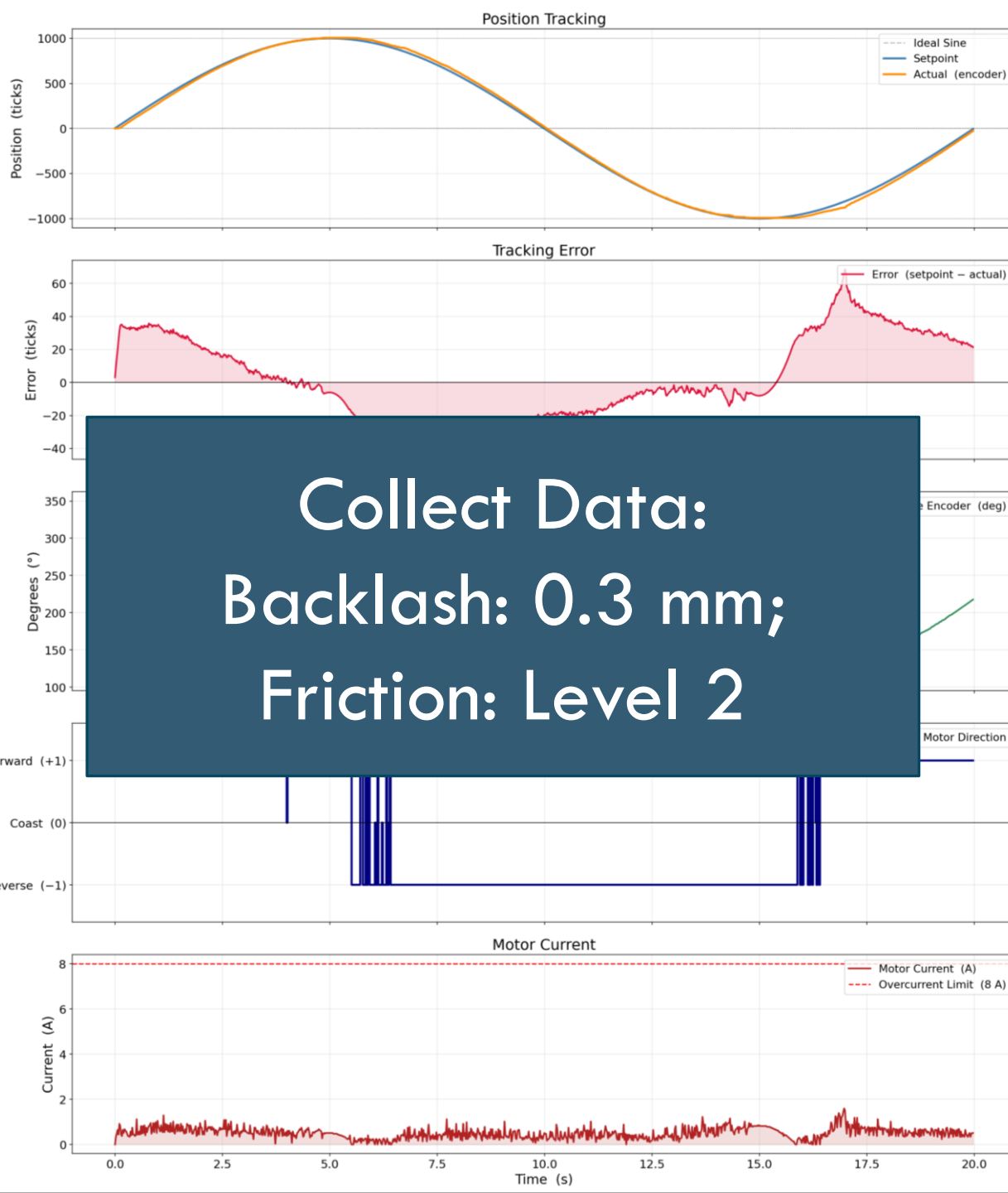
Default Backlash

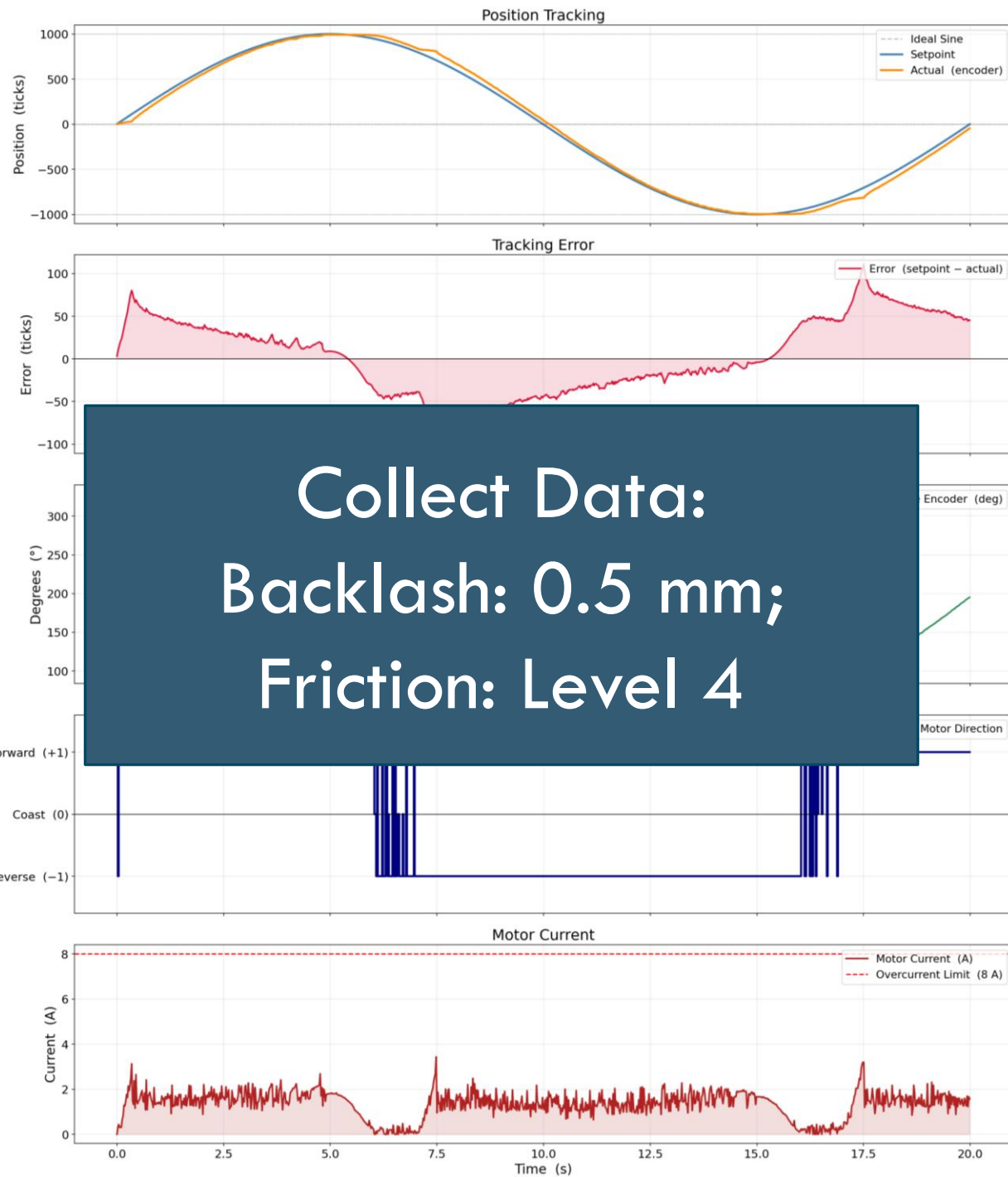


RUN EXPERIMENTS

		Backlash Levels				
	-	Default	0.1 mm	0.3 mm	0.4 mm	0.5 mm
Friction Levels	0.212 in	Trial 01	Trial 02	Trial 03	Trial 04	Trial 05
	0.205 in	Trial 06	Trial 07	Trial 08	Trial 09	Trial 10
	0.195 in	Trial 11	Trial 12	Trial 13	Trial 14	Trial 15
	0.162 in	Trial 16	Trial 17	Trial 18	Trial 19	Trial 20

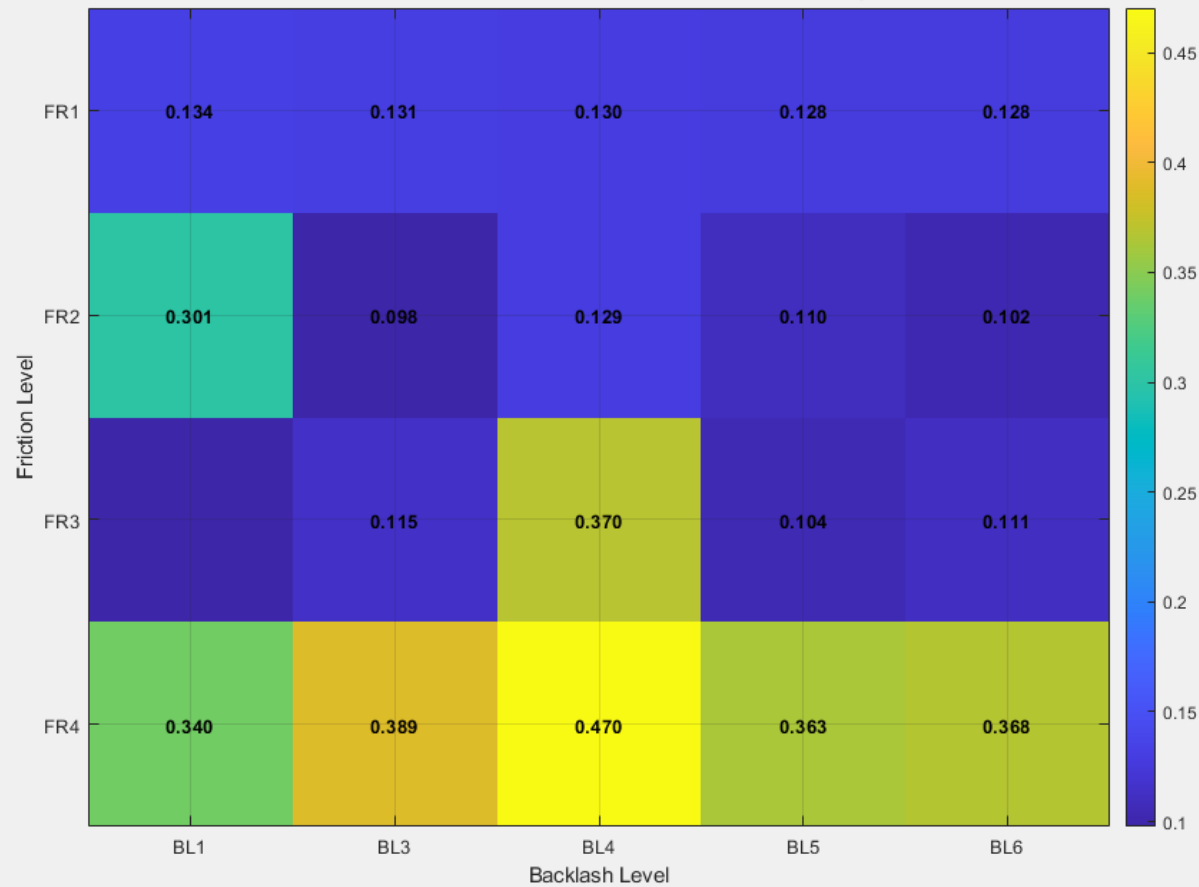




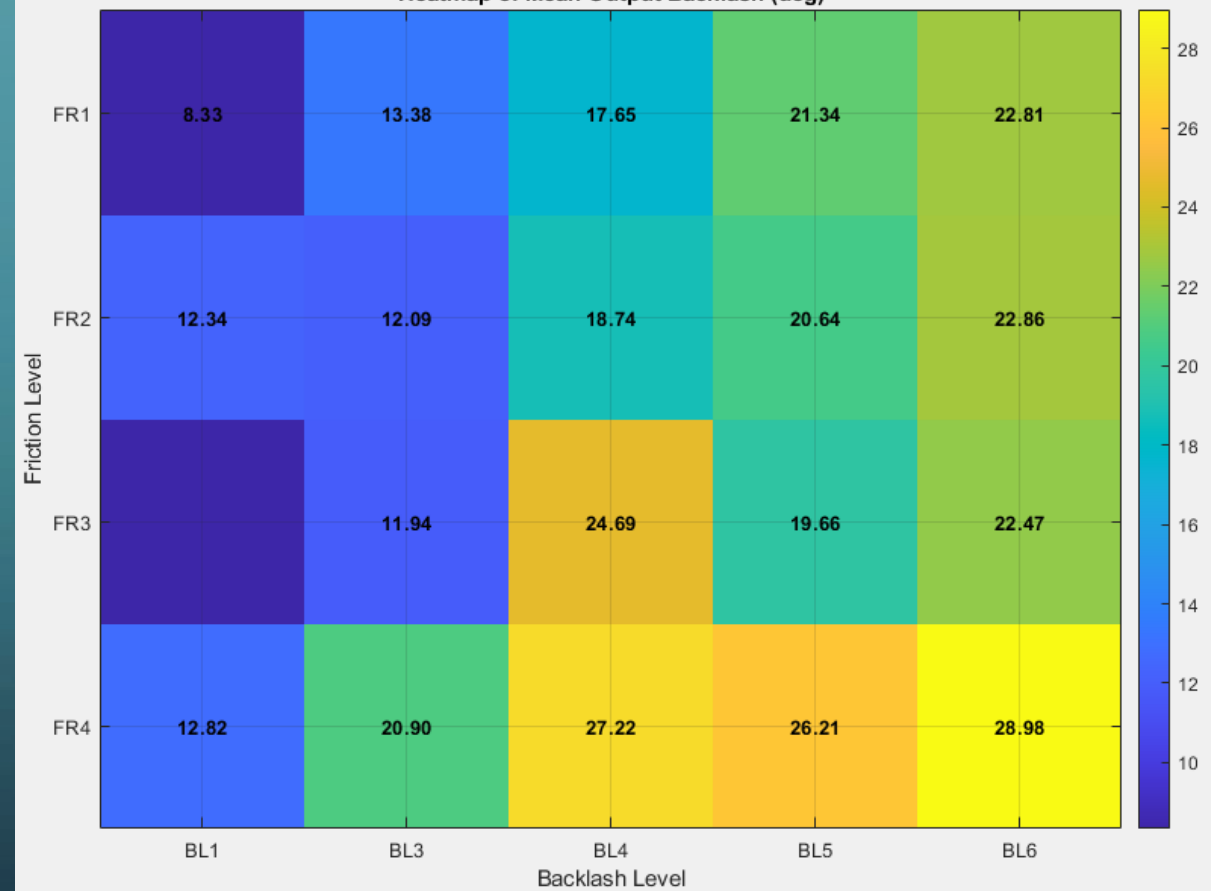


MEASURED FRICTION AND BACKLASH

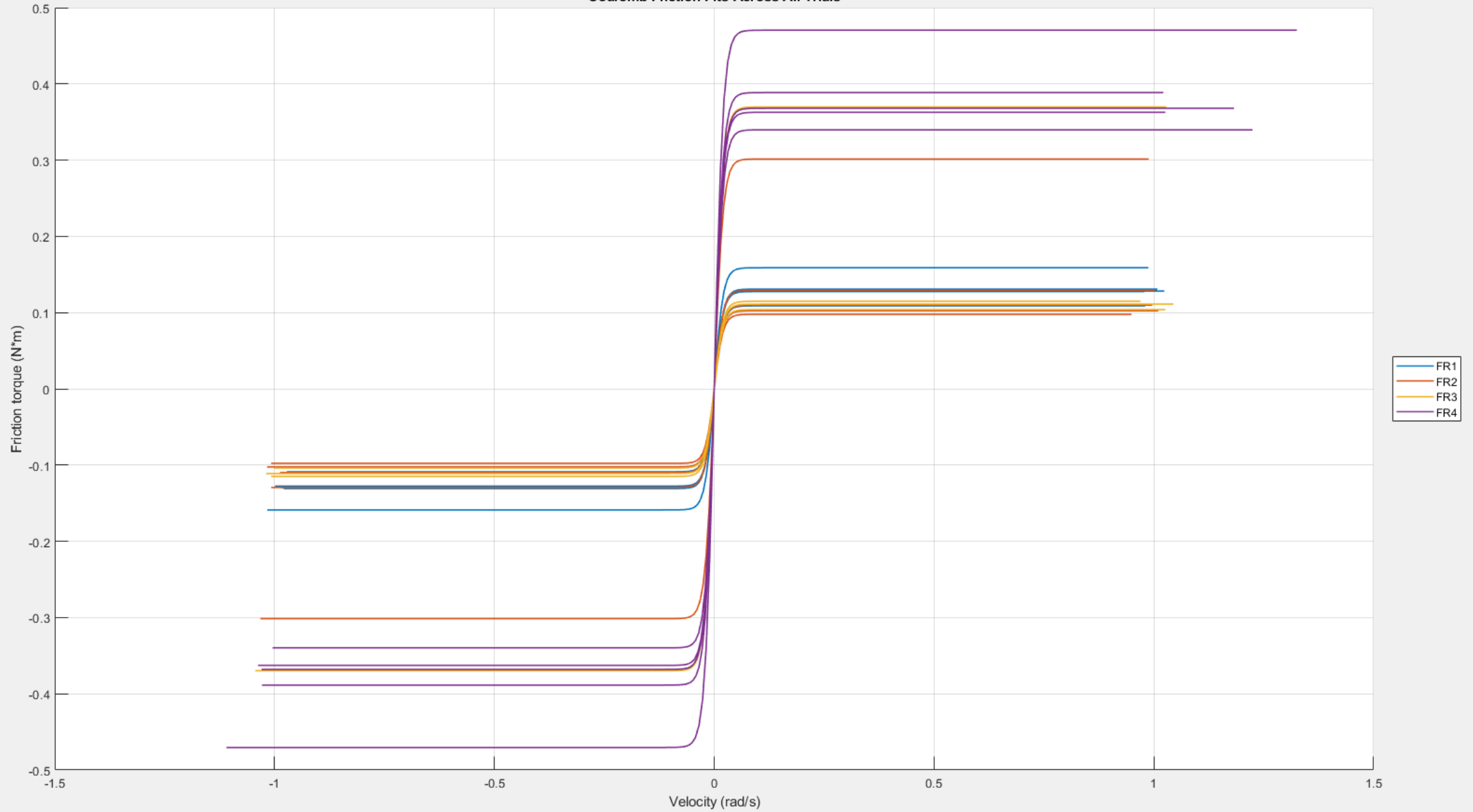
Heatmap of Identified Coulomb Friction Coefficient (F_c)



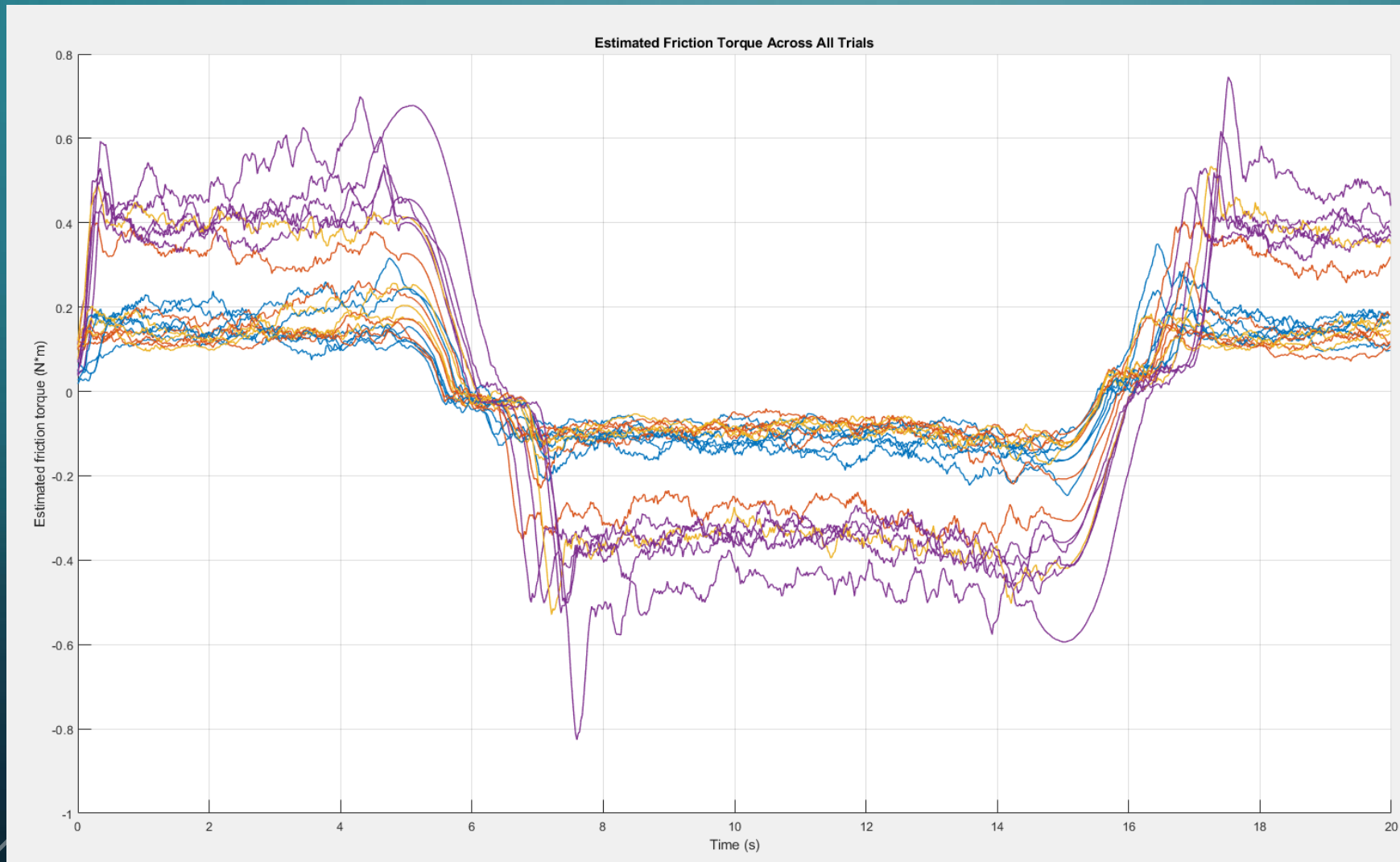
Heatmap of Mean Output Backlash (deg)



Coulomb Friction Fits Across All Trials

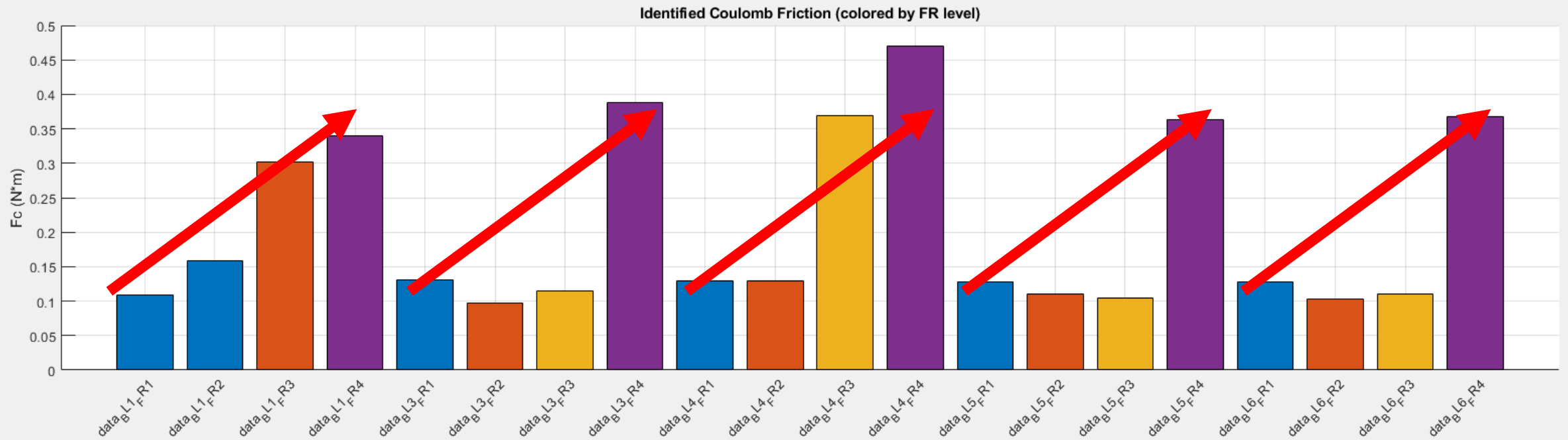


MODEL NONLINEARITIES

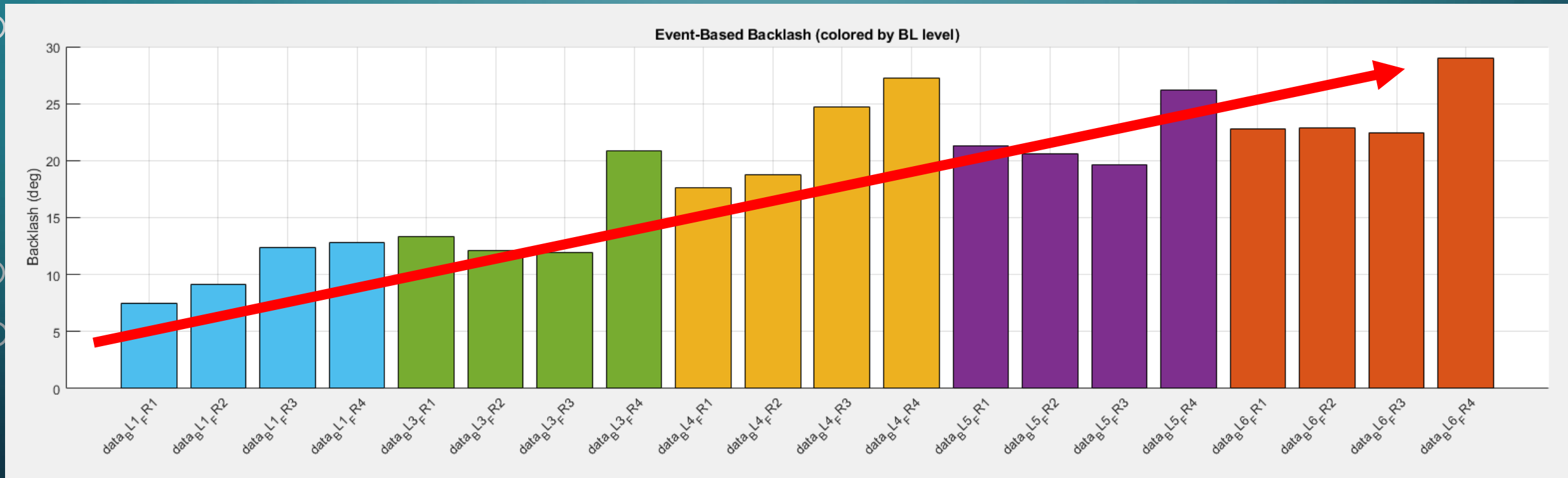


- ★ Friction Level 1
- ★ Friction Level 2
- ★ Friction Level 3
- ★ Friction Level 4

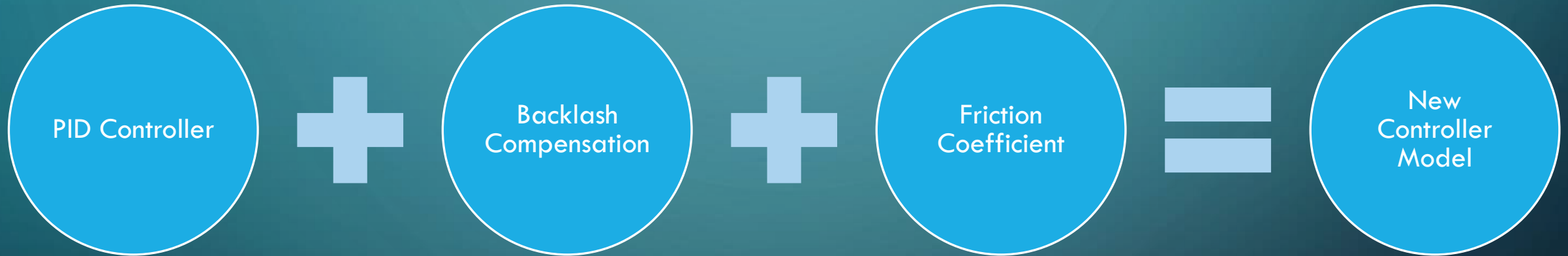
FIT PARAMETERS: FRICTION



FIT PARAMETERS: BACKLASH



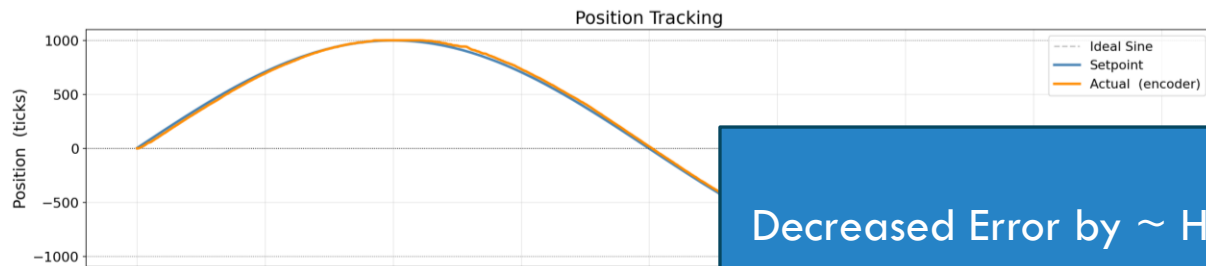
FRICTION + BACKLASH CONTROLLER



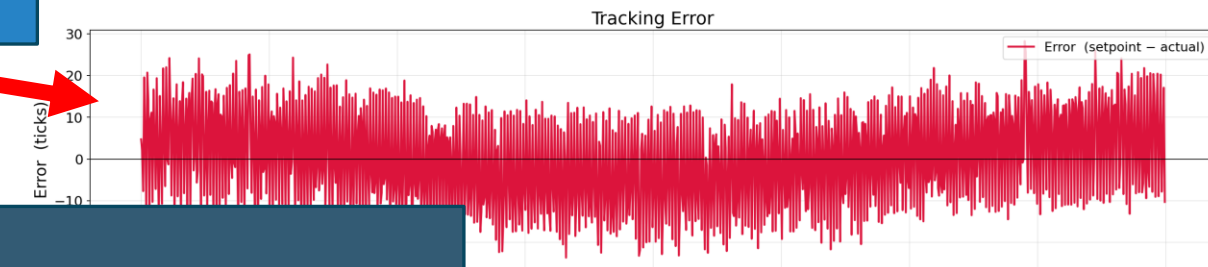
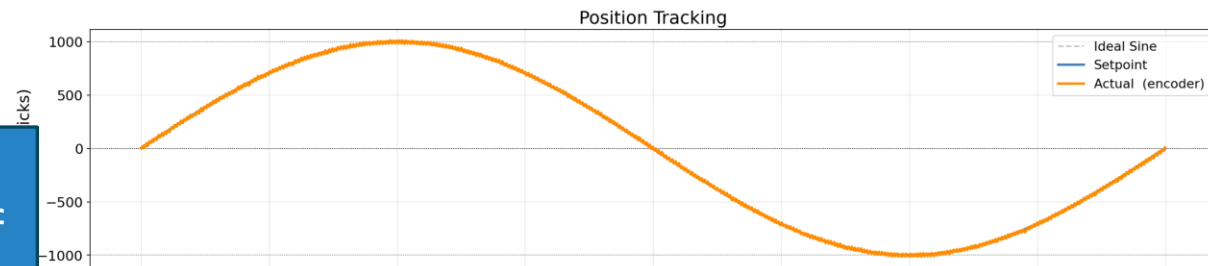
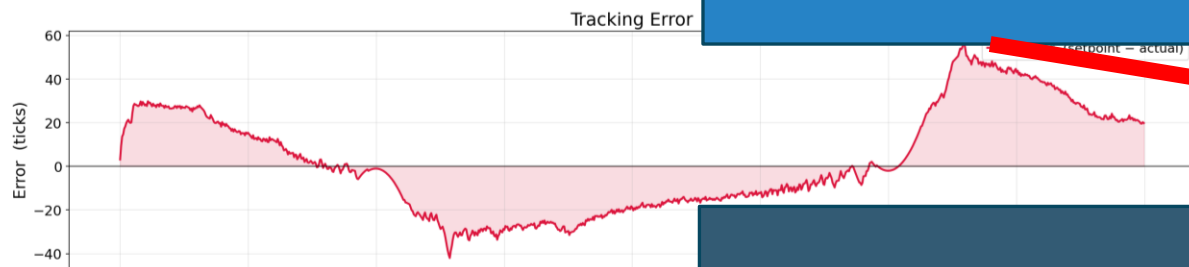
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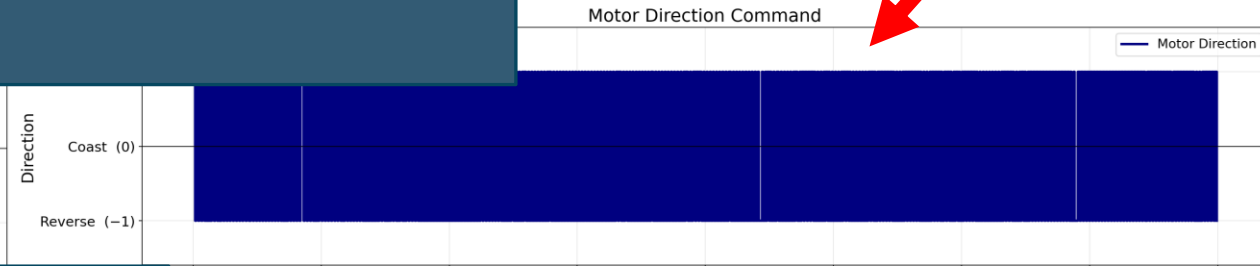
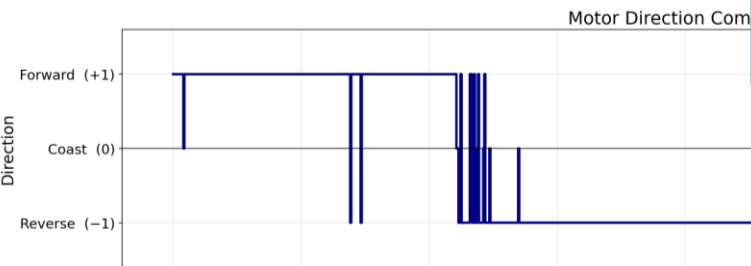
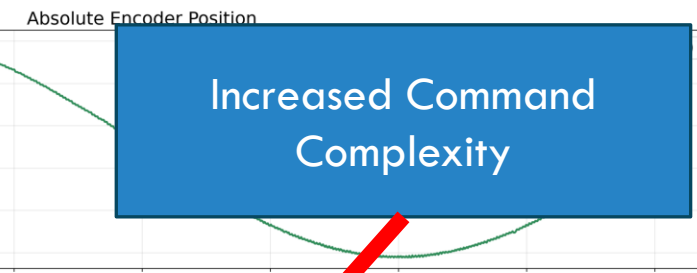
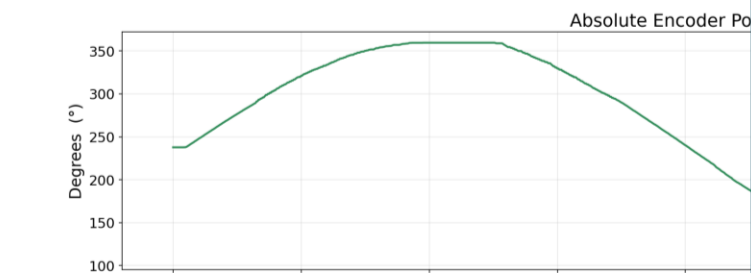


Decreased Error by ~ Half

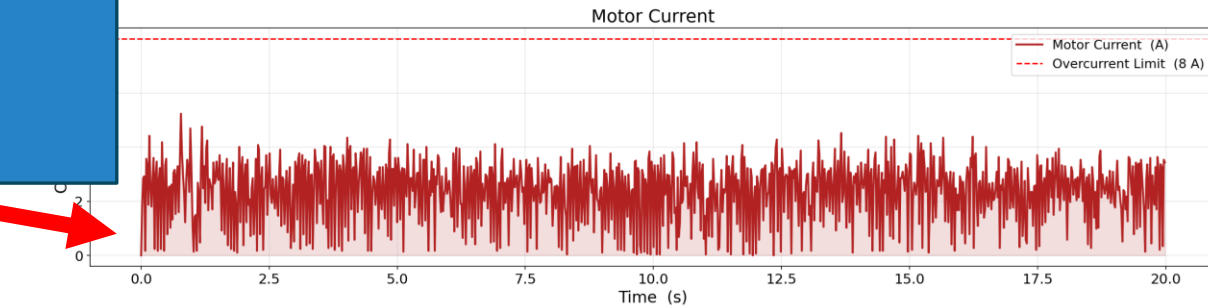
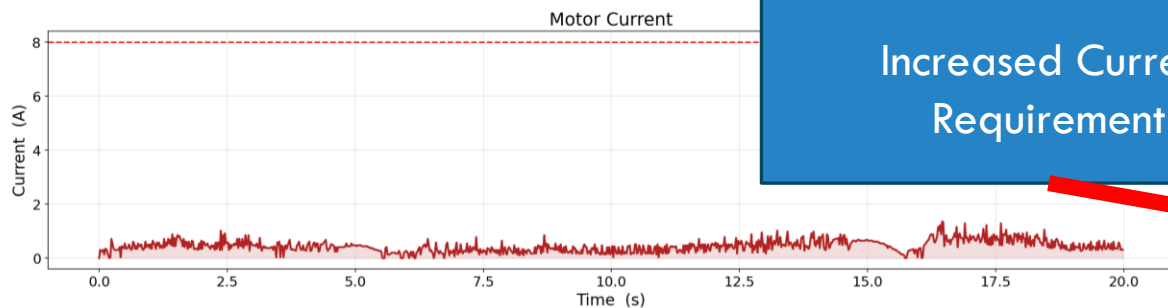


CONTROLLER VALIDATION

Increased Command
Complexity



Increased Current
Requirement



MODEL LIMITATIONS

- Friction Definition:
 - Don't define distance by rotation of screw, instead define by 4 Pin Gauges
- Current Surges
 - Current is vastly increased using this model. The motor flips back and forth in direction, so there are surges in current.
 - Previously, if there was friction it would be jerky. The motion is smoother with this controller, but the motor is also working much harder and therefore hitting these current limits.

CONCLUSION

- ✓ Friction and backlash significantly impact real robotic system performance
- ✓ Coulomb friction model captures dominant friction behavior across conditions
- ✓ Event-based method provides a practical and interpretable measure of backlash
- ✓ Experimental results show measurable variation with friction and backlash levels
- ✓ Incorporating these effects improves understanding beyond ideal kinematic models